

Procurement Analytics Dashboard

Operations & Supply Chain Analytics Project

Tools Used

- Power BI
- MySQL
- Power Query
- Excel

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Executive Summary

This project focuses on analyzing procurement and supply chain performance using the DataCo Supply Chain dataset containing 180,520 transaction records. The objective was to transform raw operational data into meaningful business insights through data cleaning, SQL analysis, and Power BI dashboard development. The dashboard enables procurement managers to monitor spending, delivery performance, and logistics efficiency using interactive visualizations.

Project Objectives

- Analyze procurement spending.
- Monitor delivery performance.
- Evaluate shipping modes.
- Identify high-spending product categories.
- Build an interactive executive dashboard.
- Support procurement decision-making.

Dataset Overview

Dataset Name: DataCo Supply Chain Dataset

Source: Kaggle

Total Records: 180,520

Total Columns: 54

Example fields:

Column	Description
Order Id	Unique order
Sales	Sales amount
Market	Sales region
Category Name	Product category
Delivery Status	Delivery performance
Shipping Mode	Shipping method

Methodology

The Procurement Analytics project followed a structured business intelligence workflow to transform raw procurement data into meaningful insights.

Step 1 – Data Collection

The **DataCo Supply Chain Dataset** containing **180,520 transaction records** and **54 attributes** was obtained for analysis. The dataset includes procurement transactions, customer information, product details, shipping information, delivery status, and regional sales data.

Step 2 – Data Understanding

The dataset structure was examined using Excel and the accompanying data dictionary. Key business fields such as Sales, Order ID, Market, Category Name, Shipping Mode, Delivery Status, Order Date, and Quantity were identified for analysis.

Step 3 – Data Cleaning and Transformation

Due to the large dataset size, data preparation was performed using **Power Query in Power BI**. The following transformations were applied:

- Verified column data types.

- Renamed columns for improved readability.
- Removed unnecessary fields from reporting.
- Checked for missing or inconsistent values.
- Created a calendar (date) table for time-based analysis.

Step 4 – Data Modeling

A dimensional model was developed in Power BI by establishing relationships between the fact table and the date dimension. This enabled efficient filtering and time-based reporting.

Step 5 – KPI Development

Key procurement performance indicators were developed using DAX measures, including:

- Total Purchase Spend
- Total Orders
- Total Quantity Purchased
- On-Time Delivery Percentage
- Average Delivery Delay

These KPIs provide a quantitative assessment of procurement and logistics performance.

Step 6 – Dashboard Development

An interactive dashboard was designed in Power BI using KPI cards, line charts, bar charts, donut charts, and slicers. The dashboard enables users to analyze procurement performance by market, category, shipping mode, and time period.

Step 7 – SQL Validation

The dataset was imported into MySQL Workbench, and SQL queries were executed to validate dashboard calculations and perform business analysis. SQL techniques included:

- Aggregate functions (SUM, COUNT, AVG)
- Filtering (WHERE)
- Grouping (GROUP BY)
- Sorting (ORDER BY)
- Views
- Ranking functions
- Common Table Expressions (CTEs)

Step 8 – Business Insight Generation

The final dashboard was analyzed to identify procurement spending patterns, regional performance, category-wise expenditure, delivery efficiency, and logistics trends. These insights can support procurement managers in making data-driven operational decisions.

Data Collection

Downloaded the DataCo Supply Chain Dataset.

Data Preparation

Performed data transformation using Power Query.

Data Modeling

Built a star schema in Power BI.

KPI Development

Created DAX measures.

SQL Analysis

Validated KPIs using MySQL.

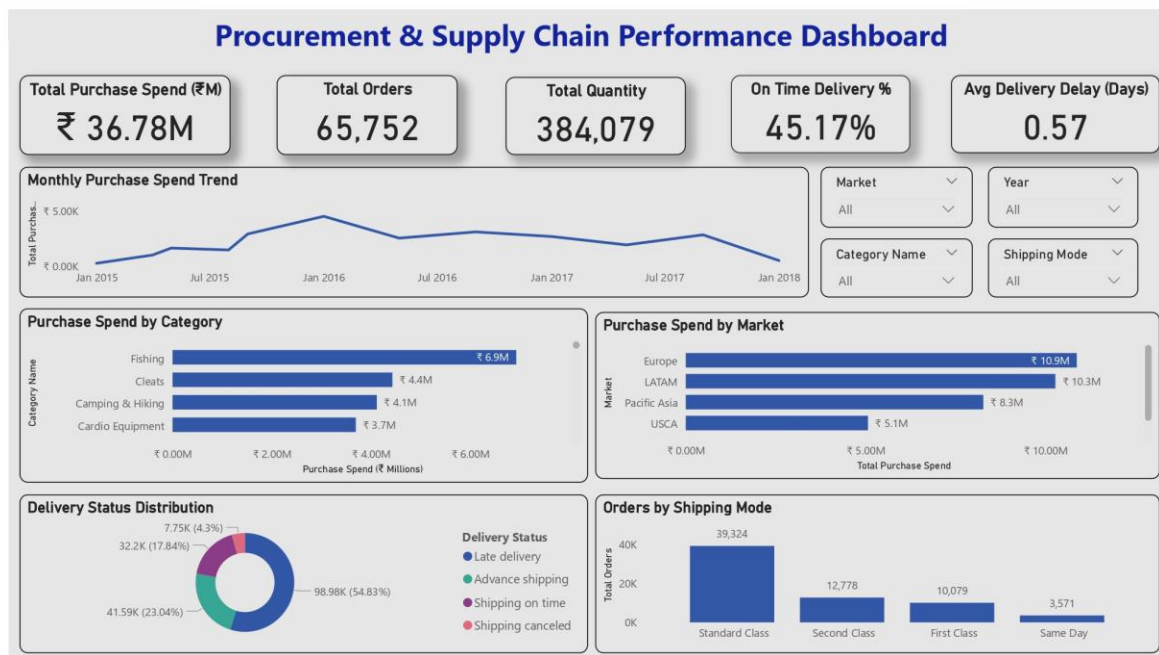
Dashboard Development

Designed an interactive executive dashboard.

Dashboard KPIs

KPI	Description
Purchase Spend	Total procurement value
Total Orders	Number of unique orders
Total Quantity	Total quantity purchased
Average Delivery Delay	Average shipping delay
On-Time Delivery %	Percentage delivered on time

Dashboard Visualizations



KPI Cards

1. Total Purchase Spend (₹36.78M)

What it shows

This KPI displays the total procurement expenditure recorded in the dataset.

Business Importance

- Measures overall purchasing activity.
- Helps management monitor procurement budgets.
- Enables comparison of spending across different time periods and markets.

Business Insight

The organization processed procurement transactions worth approximately **₹36.78 million**, indicating the overall scale of procurement operations during the analysis period.

2. Total Orders (65,752)

What it shows

Displays the total number of unique customer orders.

Business Importance

- Indicates procurement transaction volume.
- Helps evaluate operational workload.
- Supports demand analysis.

Business Insight

More than **65,000 procurement orders** were processed, reflecting a high volume of supply chain transactions.

3. Total Quantity (384,079)

What it shows

Represents the total number of products purchased across all orders.

Business Importance

- Measures procurement volume.
- Assists inventory planning.
- Supports warehouse capacity analysis.

Business Insight

The organization purchased over **384,000 units**, indicating significant procurement activity.

4. On-Time Delivery Percentage (45.17%)

What it shows

Percentage of deliveries completed on time.

Business Importance

- Evaluates logistics performance.
- Measures supplier reliability.
- Tracks supply chain efficiency.

Business Insight

Only **45.17% of deliveries were completed on time**, highlighting opportunities to improve supplier coordination and logistics planning.

5. Average Delivery Delay (0.57 Days)

What it shows

Average difference between actual and scheduled delivery days.

Business Importance

- Measures delivery efficiency.
- Helps identify logistics delays.
- Supports continuous process improvement.

Business Insight

The average delivery delay is **0.57 days**, indicating relatively small delays but still presenting opportunities for optimization.

Monthly Purchase Spend Trend

What it shows

Displays procurement spending over time (January 2015 – January 2018).

Business Importance

- Identifies spending patterns.
- Detects seasonal procurement trends.
- Supports procurement budgeting.

Interpretation

The spending trend fluctuates over the observed period, with the highest procurement expenditure occurring around **early 2016**, followed by moderate variations. Such analysis helps procurement managers anticipate purchasing requirements and allocate budgets effectively.

Purchase Spend by Category

What it shows

Displays procurement expenditure across different product categories.

Business Importance

- Identifies high-spending categories.

- Supports inventory prioritization.
- Assists supplier negotiations.

Interpretation

The **Fishing** category contributes the highest procurement expenditure (approximately **₹6.9 million**), followed by **Cleats, Camping & Hiking**, and **Cardio Equipment**. Procurement managers can focus cost optimization efforts on these categories due to their significant contribution to total spending.

Purchase Spend by Market

What it shows

Compares procurement expenditure across different geographical markets.

Business Importance

- Identifies major procurement regions.
- Supports regional supplier management.
- Assists strategic procurement planning.

Interpretation

Europe contributes the highest procurement spend (approximately **₹10.9 million**), followed closely by **LATAM** (₹10.3 million). These regions represent the largest procurement markets and should be prioritized for supplier relationship management and procurement planning.

Delivery Status Distribution

What it shows

Displays the proportion of orders by delivery status.

Business Importance

- Evaluates logistics performance.
- Monitors supplier reliability.
- Tracks delivery efficiency.

Interpretation

The largest share of deliveries falls under **Late Delivery (54.83%)**, followed by **Advance Shipping (23.04%)**, **Shipping on Time (17.84%)**, and **Shipping Cancelled (4.3%)**. The high percentage of

late deliveries indicates the need for improved logistics coordination and supplier performance monitoring.

Orders by Shipping Mode

What it shows

Displays the number of orders shipped using each shipping mode.

Business Importance

- Identifies preferred shipping methods.
- Supports logistics planning.
- Evaluates transportation utilization.

Interpretation

Standard Class is the most frequently used shipping mode with **39,324 orders**, followed by **Second Class**, **First Class**, and **Same Day** delivery. Since Standard Class handles the majority of shipments, improvements in this shipping mode can have the greatest operational impact.

Interactive Slicers

The dashboard includes four slicers:

- Market
- Year
- Category Name
- Shipping Mode

Purpose

These filters allow users to dynamically analyze procurement performance by different dimensions without modifying the dashboard design.

SQL Analysis

The SQL analysis validated the dashboard KPIs using:

- Aggregate Functions
- GROUP BY
- ORDER BY
- HAVING

- Views
- Ranking Functions

Sample Query:

```
SELECT Market,  
SUM(Sales)  
FROM Fact_Orders  
GROUP BY Market;
```

Business Insights

Insight 1

Purchase Spend

Total Purchase Spend is ₹36.78 Million.

The organization processed procurement worth ₹36.78M during the analysis period.

Insight 2

Top Market

Europe contributes the highest procurement spend.

Europe should receive strategic supplier management attention because it represents the highest spending region.

Insight 3

Top Categories

Fishing
Cleats
Camping & Hiking

These categories contribute the largest procurement expenditure.

Insight 4

Shipping Mode

Standard Class accounts for most shipments.

Logistics optimization should focus on Standard Class because operational improvements here will have the biggest impact.

Insight 5

Delivery Performance

45.17% deliveries are on time.

Delivery performance needs improvement through supplier monitoring and logistics optimization.

Recommendations

This section is very important because it shows analytical thinking.

Example:

- Monitor high-spending categories regularly.
- Improve on-time delivery through supplier performance tracking.
- Review logistics for delayed shipments.
- Use dashboard KPIs in monthly procurement review meetings.
- Continue monitoring market-wise procurement trends.

Conclusion

This project demonstrates how Power BI, SQL, and Power Query can be integrated to analyze procurement operations using a large real-world dataset. The dashboard enables procurement managers to monitor KPIs, identify trends, and make informed operational decisions.

Appendix (Optional)

Include:

- Data Dictionary
- Dashboard Screenshot
- Sample SQL Queries